

FILTER BASED NOISE SHAPING SAR-ADC FOR LOW POWER APPLICATIONS

Deepak K (Roll. No:715521106009)
Deepesh M (Roll. No:715521106010)
Kishore C (Roll. No:715521106024)
Prasanna Kumar K (Roll. No:715521106034)

Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

**Branch: ELECTRONICS AND COMMUNICATION
ENGINEERING**

of Anna University



May 2025

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**PSG INSTITUTE OF TECHNOLOGY AND APPLIED
RESEARCH**

(Autonomous Institution)

COIMBATORE – 641 062

PSG INSTITUTE OF TECHNOLOGY AND APPLIED RESEARCH

(Autonomous Institution)

COIMBATORE – 641 062

FILTER BASED NOISE SHAPING SAR-ADC FOR LOW POWER APPLICATIONS

Bonafide record of work done by

Deepak K	(715521106009)
Deepesh M	(715521106010)
Kishore C	(715521106024)
Prasanna Kumar K	(715521106034)

Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: ELECTRONICS AND COMMUNICATION ENGINEERING

of Anna University

May 2025

.....
Dr. K.Paldurai
Faculty Guide
Professor

.....
Dr.P.Vijayakumar
Head of the Department

Certified that the candidate was examined in the viva-voce examination held on

.....
(Internal Examiner)

.....
(External Examiner)

CONTENTS

CHAPTER NO.	PAGE NO.
ACKNOWLEDGEMENT.....	(v)
ABSTRACT.....	(vi)
LIST OF TABLES.....	(vii)
LIST OF FIGURES.....	(viii)
LIST OF ABBREVIATIONS AND SYMBOLS.....	(ix)
1. INTRODUCTION.....	1
1.1 NEED FOR THE PROJECT	2
1.2 PROBLEM STATEMENT	3
1.3 FILTER BASED CIRCUIT	3
2. SUCCESSIVE APPROXIMATION ADC ARCHITECTURES.....	8
2.1 CONVENTIONAL SAR ADC	8
2.1.1 WORKING	8
2.1.2 ADVANTAGES	9
2.1.3 DISADVANTAGES	9
2.2 SPLIT CAPACITOR SAR ADC	9
2.2.1 WORKING	9
2.2.2 ADVANTAGES	10
2.2.3 DISADVANTAGES	10
2.3 MONOTONIC CAPACITOR SWITCHING SAR ADC	10
2.3.1 WORKING	11
2.3.2 ADVANTAGES	11
2.3.3 DISADVANTAGES	11
2.4 CHARGE REDISTRIBUTION SAR ADC	11
2.3.1 WORKING	12
2.3.2 ADVANTAGES	12
2.3.3 DISADVANTAGES	12
2.5 TWO STEP ADC	12
2.5.1 WORKING	13
2.5.2 ADVANTAGES	13

2.5.3	DISADVANTAGES	13
2.6	TIME-INTERLEAVED SAR ADC	13
2.6.1	WORKING	14
2.6.2	ADVANTAGES	14
2.6.3	DISADVANTAGES	14
2.7	ASYNCHRONOUS SAR ADC	14
2.7.1	WORKING	15
2.7.2	ADVANTAGES	15
2.7.3	DISADVANTAGES	15
3.	LITERATURE SURVEY.....	16
4.	TOOL DESCRIPTION.....	20
4.1	CADENCE	20
5.	PROPOSED METHODOLOGY.....	24
5.1	FULLY DIFFERENTIAL OTA-BASED LOOP FILTER DESIGN	24
5.1.2	CIRCUIT IMPLEMENTATION	25
5.2	FILTER BASED SAR ADC	28
5.2.1	CIRCUIT IMPLEMENTATION	31
6.	EXPERIMENTS AND RESULTS.....	36
6.1	OTA-BASED LOOP FILTER FOR NS-SAR ADC	36
6.1.1	COMPARISON WITH EXISTING RESULTS	37
7.	CONCLUSION.....	38
	BIBLIOGRAPHY	40

ACKNOWLEDGEMENT

We are grateful to **Dr. P. V. Mohanram**, Secretary of PSG Institute of Technology and Applied Research for giving us support throughout the project.

We express our sincere thanks to **Dr. N. Saravana Kumar**, Principal, PSG Institute of Technology and Applied Research, for providing the necessary support and opportunity to carry out this project.

We extend our profound gratitude to **Dr. P. Vijayakumar**, Head of the Department, Department of Electronics and Communication Engineering, for his constant support and encouragement throughout the project. We also thank him for serving as a project committee member and rendering valuable feedback and guidance during the review sessions.

We express our heartfelt thanks to **Dr. K. Paldurai**, Professor, Department of Electronics and Communication Engineering, and our Project Guide, for his consistent support, valuable suggestions, and timely motivation at various stages of the work. His able mentorship and encouragement have been instrumental in the successful completion of this project. We are extremely grateful for the wonderful experience we had while working with him.

We would like to express our sincere thanks to all teaching and non-teaching staff members in the Department of Electronics and Communication Engineering for their direct and indirect support, which greatly contributed to the successful completion of this project.

ABSTRACT

A low-power, filter-based noise-shaping Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) is designed and implemented for high-resolution applications. Traditional SAR ADCs encounter challenges such as comparator noise, capacitive mismatch errors, and increased power dissipation, which constrain their performance in high-resolution applications. To mitigate these issues, a novel noise-shaping SAR ADC architecture incorporating a fully differential loop filter is proposed to enhance resolution while ensuring power efficiency and robustness against process-voltage-temperature (PVT) variations. The proposed design integrates a bootstrapped sampling circuit, a capacitive DAC (CDAC) for digital-to-analog conversion, and a dynamic comparator to improve decision-making accuracy. The residual voltage from the conversion process is processed through a loop filter and an integrator, enabling superior linearity and noise suppression. The architecture combines a fully differential Operational Transconductance Amplifier (OTA) with an active-RC ramp generator to enhance residue integration. The proposed ADC was analyzed theoretically, evaluated for performance, and compared with state-of-the-art designs, demonstrating superior efficiency

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
6.1	Summary of the recently published SAR ADC performance parameters	37

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
1.1	Block diagram of a SAR ADC	4
2.1	ADC Block Diagram	8
2.2	Split Capacitor SAR ADC	9
2.3	Monotonic Capacitor Switching SAR ADC	10
2.4	Charge Redistribution SAR ADC	11
2.5	Two Step SAR ADC	12
2.6	Time Interleaved SAR ADC	13
2.7	Asynchronous SAR ADC	14
4.1	An inverter cell with three views: layout, schematic, library	20
4.2	The design flow	22
5.1	Basic Differential OTA Filter	24
5.2	Fully Differential Operational Transconductance Amplifier (OTA) filter	25
5.3	Proposed Operational Transconductance Amplifier (OTA) filter	26
5.4	Flowchart of Filter Based SAR ADC	29
5.5	Input Bootstrap Sampling Switch	29
5.6	Capacitive Based DAC	30
5.7	Dynamic Clocked Comparator	31
5.8	Circuit Implementation of Input Bootstrap Sampling Switch	32
5.9	Circuit Implementation of Capacitive Based DAC	32
5.10	Circuit Implementation of Dynamic Clocked Comparator	33
5.11	Circuit Implementation of SAR Logic	33
5.12	Circuit Implementation of Filter based SAR ADC	35
6.1	OTA Filter Output Waveform	36
6.2	Output of Filter Based Noise Shaping SAR ADC	36

LIST OF ABBREVIATIONS AND SYMBOLS

VLSI	–	Very Large-Scale Integration
ADC	–	Analog to Digital Converter
DAC	–	Digital to Analog Converter
SoC	–	System on Chip
DNL	–	Differential Non-Linearity
INL	–	Integral Non-Linearity
LSB	–	Least Significant Bit
MSB	–	Most Significant Bit
S/H	–	Sample and Hold
SAR	–	Successive Approximation Register
NS-SAR	–	Noise Shaping -Successive Approximation Register
Op-Amp	–	Operational Amplifier
PVT	–	Process Voltage Temperature
CMOS	–	Complementary Metal Oxide Semiconductor
OTA	–	Operational Trans-Conductance Amplifier
MOSFET	–	Metal Oxide Semiconductor Field Effect Transistor
DFW	–	Design Framework
CIW	–	Command Interpreter Window
FIR	–	Finite Impulse Response
IIR	–	Infinite Impulse Response
DW	–	Design Window
TW	–	Text Window
DRC	–	Design Rule Check
HDL	–	Hardware Description Language
ISE	–	Integrated Synthesis Environment
ns	–	Nanoseconds
μs	–	Microsecond
ENOB	–	Effective Number of Bits
SNDR	–	Signal-to-Noise and Distortion Ratio
SFDR	–	Spurious-Free Dynamic Range

CHAPTER 1

INTRODUCTION

Electronics have become an integral part of our daily lives. The rapid growth of electronics in recent decades has made our lives more convenient, safe, and secure. It might be impossible to visualize a world without electronics. Everything from cooking to laundry to exercise to music involves electronics in some way. Electronics is widely used in communication, military, entertainment, medical, industrial automation, and instrumentation. Electronics enables the automation of complex tasks, fast computing, and high-speed data transmission, ultimately increasing yield and accuracy.

With the invention of the transistor in 1947, the later half of the century witnessed rapid growth in solid state electronics. Though the concept of Integrated Circuits were introduced in 1958, it was in 1970's they were commercialized. An integrated circuit is a group of electronic devices, both active and passive, on a single compact flat piece of semiconductor material. Gordon Moore, an American engineer, in 1965, proposed his observation that "the number of transistors per unit area in an integrated circuit doubles every 18 months" and the prediction is still holding true.

The advancements in integrated circuit technologies are primarily responsible for the electronics industry's extraordinary growth. Hundreds of thousands of transistors are packed onto a single silicon chip with VLSI integrated circuit technology. Due to VLSI technology, today's devices such as smartphones and cellular communications provide exceptional portability, processing capacity, and application access.

Analog and digital electronic circuits are the two broad categories of electronic circuits. The analog circuits take analog signals as input, process them and provide analog signals as output. On the other hand, digital circuits deal only with digital signals. Unlike analog circuits, digital circuits need converters as all the real-world signals are analog. Data converters establish a link between the analog and digital domains. As a gateway between the "real world" analog domain (i.e., Voice, Video data) and the digital world of 1's and 0's, data converters have become the critical elements in the field of modern signal processing.

Analog to Digital converters and Digital to Analog converters are the two types of data converters. Analog to Digital Converter is the interface circuit which converts analog input signals to digital output signals. The Digital to Analog Converter is the interfacing circuit that transforms a digital input signal into an analog output signal. Among the core components

driving this progress, data converters play a crucial role in bridging the gap between analog and digital domains. In signal processing, communication systems, and sensor interfaces, Analog-to-Digital Converters (ADCs) and Digital-to-Analog Converters (DACs) are indispensable. These mixed-signal circuits are essential for interfacing real-world analog signals with digital processing units.

Success in achieving high precision and power efficiency in ADCs has been a primary focus in mixed-signal integrated circuit design. Among various ADC architectures, the Successive Approximation Register (SAR) ADC is widely used due to its low power consumption, moderate resolution, and efficient implementation. However, conventional SAR ADCs are limited in resolution due to the presence of quantization noise. To overcome this limitation, noise-shaping techniques have been introduced, significantly improving the Signal-to-Noise Ratio (SNR) and overall performance.

A Filter-Based Noise-Shaping SAR ADC is an advanced ADC architecture that incorporates a noise-shaping loop with filtering techniques to mitigate quantization noise and enhance resolution. By introducing a feedback filter, the quantization noise is spectrally shaped, pushing the noise to higher frequencies where it can be easily filtered out. This enables a high effective number of bits (ENOB) without the need for excessive oversampling, making it an optimal solution for low-power, high-resolution applications such as biomedical signal processing, wireless communication, and sensor interfaces.

1.1 NEED FOR THE PROJECT

Successive Approximation Register (SAR) Analog-to-Digital Converters (ADCs) are widely used in applications requiring low power consumption, high resolution, and moderate sampling rates. However, conventional SAR ADCs face limitations in achieving high resolution without significantly increasing power consumption and complexity. Noise shaping techniques have emerged as a promising solution to enhance the effective resolution of SAR ADCs by pushing quantization noise to higher frequencies, where it can be filtered out. The incorporation of a filter-based noise-shaping mechanism further improves the performance by effectively suppressing quantization noise while maintaining a simple and energy-efficient architecture. This project focuses on designing a Filter-Based Noise-Shaping SAR ADC, which integrates a filtering technique to improve noise shaping, thereby enhancing the ADC's resolution and performance. By implementing an optimized filter, this design aims to achieve better Signal-to-Noise Ratio (SNR) and reduce in-band quantization noise, making it suitable for high-precision data acquisition systems.

1.2 PROBLEM STATEMENT

With the increasing demand for high-resolution (>10 -bit) Successive Approximation Register (SAR) Analog-to-Digital Converters (ADCs) in modern electronic applications, achieving both accuracy and efficiency has become a significant challenge. Traditional SAR ADCs face performance limitations due to comparator noise, which introduces uncertainty in decision-making, and capacitive mismatch errors, which degrade linearity and overall resolution. These issues reduce precision and necessitate additional calibration techniques, increasing circuit complexity and power consumption. To address these limitations, Noise-Shaping SAR (NS SAR) ADCs have been introduced, utilizing a loop filter to suppress quantization noise and enhance resolution. While effective, this approach introduces several challenges. The integration of a noise-shaping loop leads to higher power dissipation, making it difficult to meet the stringent power constraints of battery-operated and low-power applications. Additionally, the increased circuit complexity affects the scalability of the design, making integration with System-on-Chip (SoC) architectures more challenging. Another critical issue in NS SAR ADC design is its sensitivity to process, voltage, and temperature (PVT) variations. Fluctuations in manufacturing processes, supply voltage, and temperature can significantly impact the stability and accuracy of the ADC, leading to inconsistencies in performance across different operating conditions. Ensuring robustness against PVT variations is essential for maintaining reliable performance in real-world applications. Given these challenges, there is a pressing need to develop an optimized Filter-Based Noise-Shaping SAR ADC that can achieve high resolution while maintaining low power consumption, scalability, and robustness to environmental variations. Addressing these issues will enable the development of high-performance ADCs suitable for next-generation mixed-signal applications, including biomedical signal processing, wireless communication, and sensor interfaces.

1.3 FILTER BASED CIRCUIT

In the rapidly evolving landscape of digital and mixed-signal electronics, Analog-to-Digital Converters (ADCs) play a foundational role in bridging the gap between the analog world and digital processing. The need for precise, high-resolution, and low-power data conversion is more critical than ever, particularly in domains like biomedical instrumentation, environmental sensing, industrial automation, and Internet of Things (IoT) devices. Among the various ADC architectures available, the Successive Approximation Register (SAR) ADC is particularly favored due to its relatively simple structure, deterministic operation, and high energy efficiency. These attributes make it ideal for applications that operate under power and area

constraints. However, when the resolution requirement increases—often to 14 bits or more—the traditional SAR ADC architecture begins to face substantial challenges that limit its effectiveness. Addressing these limitations without compromising the inherent strengths of the SAR ADC is an area of active research and innovation.

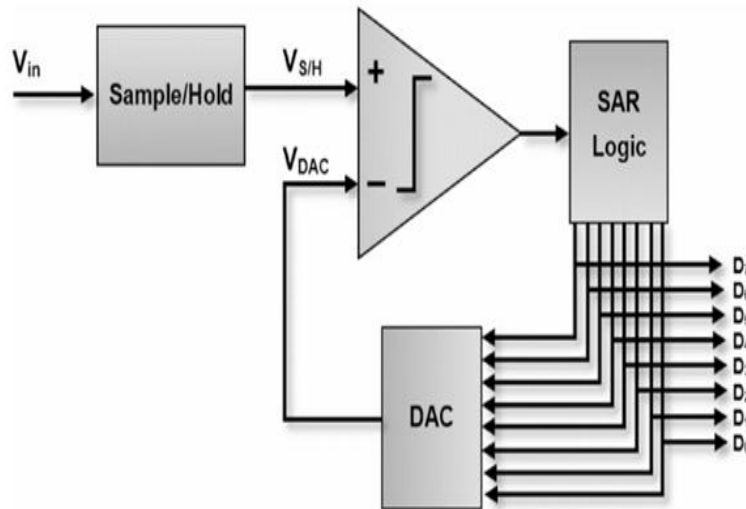


Figure 1.1 Block diagram of a SAR ADC

At its core, a SAR ADC converts an analog input signal to a digital output by performing a binary search. The architecture comprises a sample-and-hold circuit, a Digital-to-Analog Converter (DAC), a comparator, and a digital control logic block (SAR logic). The sample-and-hold circuit captures the analog input, while the DAC generates reference voltages corresponding to various digital codes. The comparator then evaluates the input against these DAC outputs under the control of SAR logic, progressively narrowing down the digital approximation over N clock cycles for an N -bit resolution. This stepwise refinement process ensures a consistent and power-efficient operation, making SAR ADCs ideal for low-to-medium resolution applications. Their implementation is also highly compatible with advanced CMOS technologies, facilitating integration in system-on-chip (SoC) designs.

However, as SAR ADCs are pushed toward higher resolution performance, several intrinsic limitations become more pronounced. One of the most fundamental issues is quantization noise, which results from the finite resolution of the digital output. In an ideal ADC, quantization noise is uniformly distributed and uncorrelated with the input, leading to a signal-to-noise ratio (SNR) that is inversely proportional to the least significant bit (LSB). For higher resolutions, the quantization step size decreases, and even small errors or mismatches in the analog components—such as capacitor mismatches in the DAC or offset in the comparator—

can lead to significant non-linearity and degradation in effective resolution. Furthermore, thermal and flicker noise, often originating from circuit elements such as switches and amplifiers, can further obscure weak analog signals, particularly in low-frequency applications like biomedical signal acquisition.

To address these challenges while maintaining the low-power, compact nature of SAR ADCs, researchers and designers have increasingly turned to filter-based circuit techniques. These techniques, particularly those involving loop filters or noise-shaping filters, have proven effective in spectrally shaping the quantization noise and pushing it out of the signal band. This approach draws inspiration from oversampled ADC architectures such as Delta-Sigma ($\Delta\Sigma$) converters, which are known for their exceptional resolution and noise performance in low-frequency domains. By integrating similar filtering principles within the SAR ADC architecture, it is possible to achieve significant improvements in performance without fundamentally altering the SAR's structural simplicity.

The basic idea behind noise shaping is to modify the frequency spectrum of the quantization noise so that it is minimized within the bandwidth of interest and instead concentrated in higher frequency regions, where it can be more easily removed using digital filters. This is typically achieved by introducing a feedback or feedforward loop with an integrator or filter that shapes the quantization error across successive conversions. For example, a first-order noise shaping loop can be introduced by accumulating the quantization error from the previous conversion and injecting it into the input of the current conversion. Higher-order noise shaping can be achieved by using multiple integrators or filters in cascade. These noise-shaping filters are implemented using either analog or digital circuits, with digital implementations offering the advantage of design flexibility and immunity to analog non-idealities.

The benefit of noise shaping is particularly evident when combined with oversampling, a technique where the ADC samples the input signal at a rate significantly higher than the Nyquist rate. Oversampling spreads the quantization noise over a broader frequency range, reducing its density in the signal band. When this is combined with noise shaping, the in-band noise can be drastically reduced, allowing for higher effective resolution without increasing the number of bits in the core ADC. This improvement is quantified by the Effective Number of Bits (ENOB), a measure of the true resolution of the converter when noise and non-linearities are taken into account. Through careful design of the noise-shaping filter and oversampling ratio, SAR ADCs can achieve ENOB values approaching those of more complex and power-hungry ADC architectures.

After the oversampled and noise-shaped output is generated, it typically undergoes digital decimation and filtering. Decimation reduces the sampling rate to the desired output data rate, while digital filters remove out-of-band noise and artifacts. One of the key advantages of this approach is that much of the precision and complexity are shifted from the analog to the digital domain. Digital circuits, especially in advanced CMOS technologies, are generally more robust to process, voltage, and temperature (PVT) variations than analog circuits. Moreover, they scale better with technology, consume less area, and offer higher design flexibility. This digital dominance aligns well with modern SoC designs, where minimizing analog components is often a key design goal.

There are several architectural variations and techniques that have been explored in the literature and industry to implement filter-based enhancements in SAR ADCs. One common approach is the use of a loop-embedded SAR ADC, where an error feedback loop is introduced around the core SAR engine. This feedback loop can be implemented using switched-capacitor circuits, current-mode circuits, or even digital filters depending on the application requirements. Another popular technique is the multi-bit quantization within each SAR cycle, where partial results are accumulated and processed over time to improve resolution. In some designs, incremental SAR ADCs are used, where the same SAR engine is reused multiple times per conversion, with filtering applied to the accumulated results.

These innovations are not just theoretical; they have been successfully demonstrated in several real-world applications. For example, in biomedical sensing, where the signals are often very low frequency and low amplitude, filter-enhanced SAR ADCs have proven effective in capturing vital parameters such as ECG or EEG with high accuracy and low power consumption. Similarly, in IoT edge devices, where energy efficiency and battery life are paramount, integrating high-resolution data converters with digital filtering allows for accurate environmental monitoring and data acquisition without draining the power budget. In industrial applications, where high precision and robustness are necessary, such ADCs enable fine-grained control and monitoring in automation systems.

Despite these advantages, implementing filter-based SAR ADCs also introduces certain design trade-offs and challenges. The addition of noise-shaping loops or oversampling control logic increases the overall system complexity and may introduce latency due to the need for multiple conversion cycles and post-processing. Moreover, designing stable and linear noise-shaping filters—especially in analog form—requires careful attention to loop dynamics, gain stability, and component matching. In digital implementations, managing dynamic range and quantization effects in intermediate stages becomes critical to prevent data overflow or loss of

precision. Therefore, the design of such systems requires a deep understanding of both analog and digital signal processing principles, as well as careful system-level planning.

From a design methodology perspective, simulation and verification of filter-based SAR ADCs are more involved than traditional SAR architectures. Designers must perform comprehensive behavioral modeling using tools like MATLAB/Simulink or SystemVerilog-AMS to validate the noise shaping performance, signal-to-noise ratio, and spectral characteristics before proceeding to transistor-level implementation. Post-layout simulations are essential to ensure that parasitics, mismatches, and non-idealities do not degrade the expected performance. Calibration and correction techniques may also be employed in the digital domain to compensate for residual errors or drift, further enhancing robustness.

In conclusion, the integration of filter-based circuit techniques into SAR ADC architectures represents a powerful and versatile solution to the challenges of high-resolution data conversion. By spectrally shaping quantization noise and redistributing it outside the signal band, these techniques enable significant improvements in effective resolution and dynamic range without fundamentally altering the SAR ADC's compact and low-power structure. The synergy of oversampling, digital decimation, and noise shaping provides a path toward high-performance, scalable, and energy-efficient ADC designs that are well-suited for the demanding requirements of modern applications. As technology continues to advance and the need for precision sensing grows, such innovations in ADC architecture will play a pivotal role in enabling the next generation of intelligent, responsive, and energy-aware electronic systems.

CHAPTER 2

SUCCESSIVE APPROXIMATION ADC ARCHITECTURES

An Analog to Digital Converter is a device that converts an analog signal into a digital signal (ADC). a digital signal is represented using binary code, which is the combination of bits 0 and 1.

The block diagram of an Analog to Digital converter is shown as below.

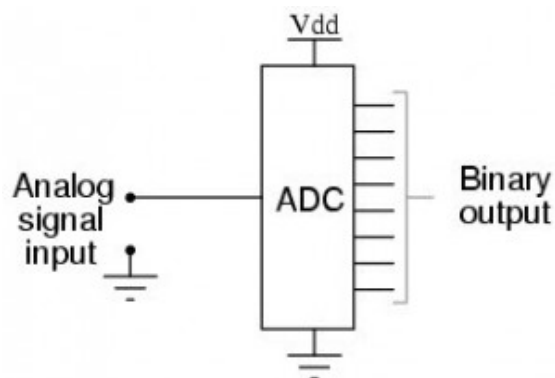


Figure 2.1 ADC Block Diagram

The Analog to Digital Converter (ADC) contains one analog signal as input and multiple digital binary signals as output, which are commonly in powers of two, as illustrated in the diagram above. The Block diagram is shown in Figure 2.1.

2.1 CONVENTIONAL SAR ADC

Conventional SAR ADC uses a binary-weighted capacitor DAC and performs conversion by successively approximating the input voltage in N steps for N-bit output.

2.1.1 WORKING

- It uses a sample-and-hold circuit, binary-weighted capacitor array, comparator, and Successive Approximation Register (SAR).
- The MSB is set to 1 and the DAC generates a voltage based on that.
- Comparator compares DAC output with input voltage, and SAR keeps or resets the bit accordingly.

- The process continues from MSB to LSB.
- After N steps, the SAR holds the binary code representing the input voltage.

2.1.2 ADVANTAGES

- Simple design and widely used in medium-resolution applications.
- Only one comparator was needed.
- Power and area-efficient.

2.1.3 DISADVANTAGES

- Slower than Flash ADC due to sequential bit decisions.
- Requires accurate capacitors, especially at higher resolutions.
- Power increases with DAC switching.

2.2 SPLIT-CAPACITOR SAR ADC

Split-Capacitor SAR ADC is a variant of the conventional SAR that splits the capacitor array into MSB and LSB sections with a bridge capacitor to reduce total capacitance. The Architecture of Split-Capacitor SAR ADC is shown in figure 2.2.

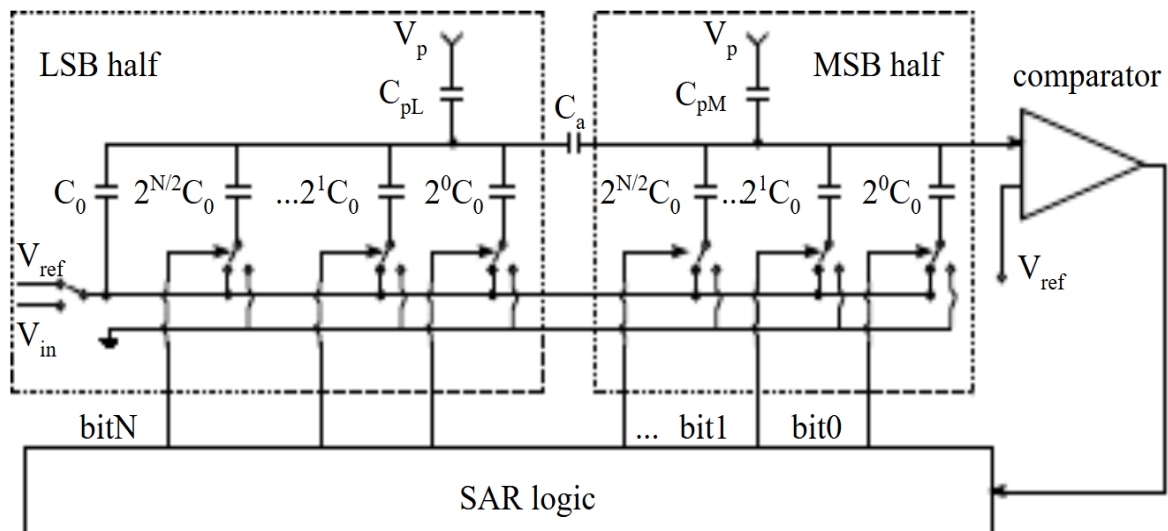


Figure 2.2 Split Capacitor SAR ADC

2.2.1 WORKING

- The capacitor DAC is divided into two sections: one for MSB and another for LSB.
- A bridge capacitor links the two parts.
- This structure reduces total capacitor size.
- During conversion, each section is used to resolve respective bits, and final output is combined.

2.2.2 ADVANTAGES

- Reduces capacitance, hence power and chip area.
- Better suited for higher resolutions than the conventional SAR.

2.2.3 DISADVANTAGES

- Requires **calibration** for accurate performance.
- Complexity increases due to segmented DAC structure.

2.3 MONOTONIC CAPACITOR SWITCHING SAR ADC

Monotonic Switching SAR ADC uses an energy-efficient capacitor switching scheme that reduces power while maintaining linearity. The Architecture of Monotonic Switching SAR ADC is shown in figure 2.3.

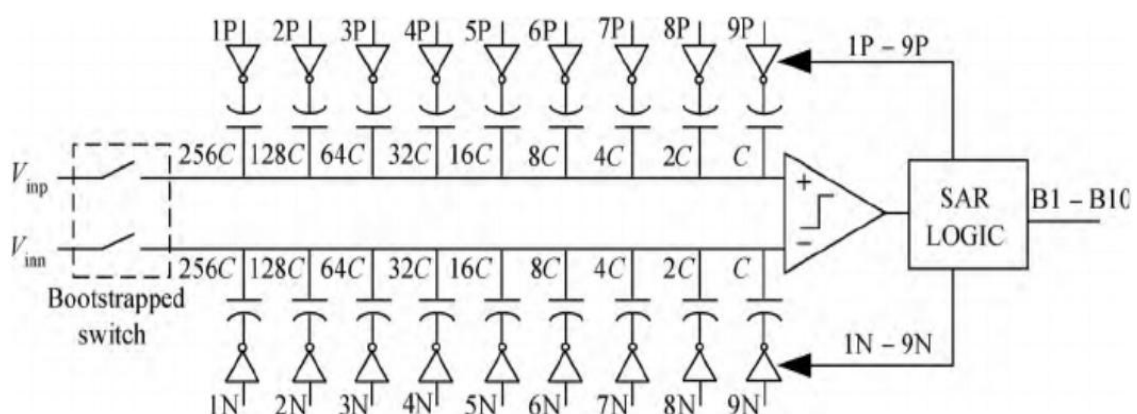


Figure 2.3 Monotonic Capacitor Switching SAR ADC

2.3.1 WORKING

- All capacitors start charged to a common mode voltage.
- Only one capacitor is switched per bit decision.
- This causes the DAC output to change monotonically, ensuring better energy efficiency.
- SAR logic records comparator output and continues until LSB.

2.3.2 ADVANTAGE

- Significantly reduces DAC switching energy (up to 87%).
- Better linearity and simplified reference voltage handling.

2.3.3 DISADVANTAGES

- Requires a more complex switching mechanism.
- Calibration may be necessary.

2.4 CHARGE REDISTRIBUTION SAR ADC

Charge Redistribution SAR ADC uses capacitors to transfer charge instead of generating voltage directly, leading to ultra-low power consumption. The Architecture of Charge Redistribution SAR ADC is shown in figure 2.4.

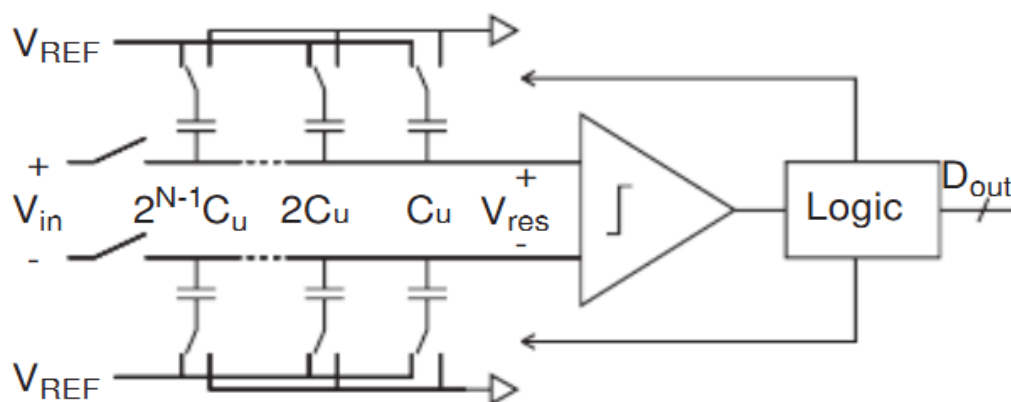


Figure 2.4 Charge Redistribution SAR ADC

2.4.1 WORKING

- During conversion, capacitors are redistributed to represent binary-weighted charge.
- Comparisons are done by measuring voltage changes due to charge redistribution.
- SAR updates bits accordingly.

2.4.2 ADVANTAGES

- Very low power consumption.
- Simple circuit architecture, especially for low resolutions.

2.4.3 DISADVANTAGES

- Prone to charge leakage and temperature variations.
- Accuracy decreases at higher resolutions.

2.5 TWO-STEP SAR ADC

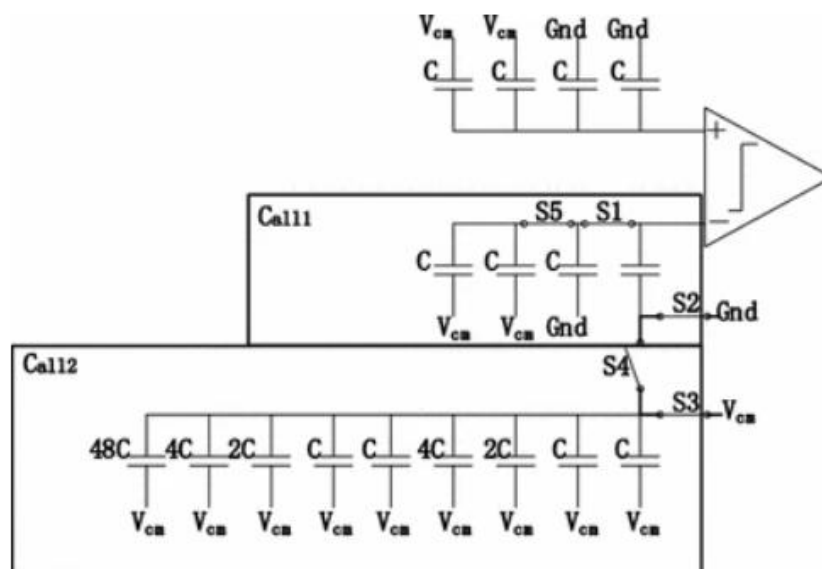


Figure 2.5 Two Step SAR ADC

Two-Step SAR ADC breaks the conversion into two phases (coarse and fine) to achieve high resolution with reduced comparison time. The Architecture of Two-Step SAR ADC is shown in figure 2.5.

2.5.1 WORKING

- The input voltage is first processed by a coarse ADC to resolve MSBs.
- Then, the residual is fed to a fine ADC to resolve LSBs.
- The final output is a combination of both stages.

2.5.2 ADVANTAGES

- Faster than single-stage SAR for high-resolution conversions.
- Reduces total number of SAR steps.

2.5.3 DISADVANTAGES

- Requires complex calibration to align both stages.
- May introduce conversion errors at stage transitions.

2.6 TIME-INTERLEAVED SAR ADC

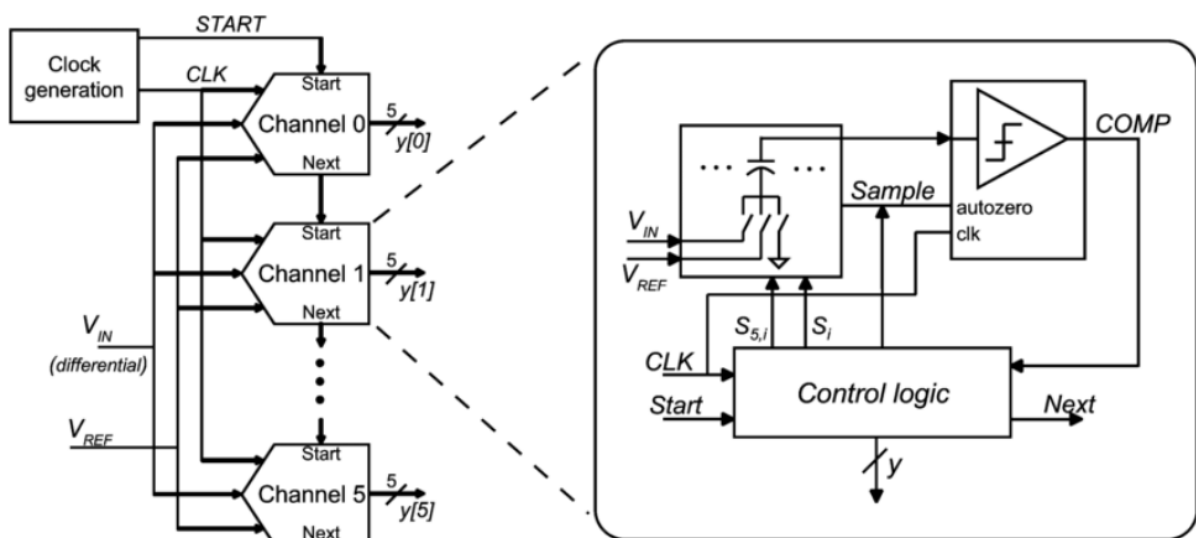


Figure 2.6 Time Interleaved SAR ADC

Time-Interleaved SAR ADC uses multiple SAR ADCs running in parallel to achieve higher effective sampling rates. The Architecture of Time-Interleaved SAR ADC is shown in figure 2.6.

2.6.1 WORKING

- Multiple SAR ADC cores are clocked in a phase-shifted manner.
- Each core samples at a different time but shares the total input bandwidth.
- The results are merged to form a high-speed output stream.

2.6.2 ADVANTAGES

- Enables very high sampling rates.
- Efficient for high-speed, high-resolution systems.

2.6.3 DISADVANTAGES

- Needs precise matching and calibration across all channels.
- Susceptible to timing errors and mismatches.

2.7 ASYNCHRONOUS SAR ADC

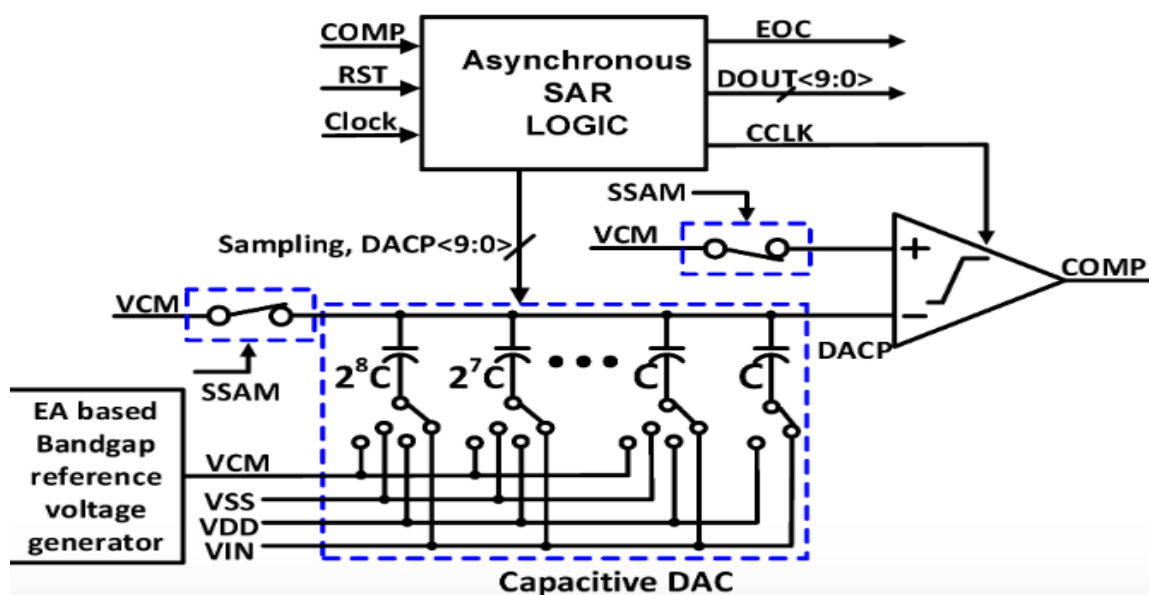


Figure 2.7 Asynchronous SAR ADC

Asynchronous SAR ADC removes the global clock and uses self-timed logic for faster and more efficient conversions. The Architecture of Asynchronous SAR ADC is shown in figure 2.7.

2.7.1 WORKING

- Each bit decision triggers the next without waiting for a global clock.
- Reduces unnecessary waiting time between SAR steps.
- Comparator, DAC, and SAR logic operate in a handshake-based fashion.

2.7.2 ADVANTAGES

- Faster than synchronous SAR ADC.
- Reduces dynamic power by avoiding clock toggling.
- Better efficiency at low voltages.

2.7.3 DISADVANTAGES

- Difficult to design and debug due to asynchronous control.
- Needs specialized timing circuits.
- Less commonly implemented in commercial ADCs.

CHAPTER 3

LITERATURE SURVEY

Accurate and efficient analog-to-digital conversion is fundamental to modern electronics, with applications ranging from communication systems to signal processing and instrumentation. Ensuring the performance of ADCs requires precise characterization and testing, as well as innovative design methodologies that balance power, area, speed, and accuracy. Two critical aspects of this domain are (1) the generation of highly linear voltage ramp signals used in ADC testing and Built-In Self-Test (BIST) circuits, and (2) advancements in ADC architectures, notably the noise-shaping successive approximation register (NS SAR) ADCs that seek to combine high resolution and low power consumption.

The characterization of ADCs often involves the measurement of static non-linearity errors such as Differential Non-Linearity (DNL) and Integral Non-Linearity (INL). A common approach to evaluating these parameters is to apply a highly linear ramp voltage input to the ADC and analyze the output codes. The accuracy of this method hinges on the linearity and stability of the ramp generator itself. Traditional ramp signal generators typically use constant current sources to charge capacitors, producing a voltage ramp proportional to time. However, these conventional designs face significant challenges.

Firstly, constant current sources based on low compliance voltage cascode current mirrors exhibit finite output impedance and leakage currents, which distort the ramp slope and degrade linearity. Moreover, designing precise current sources increases silicon area and power consumption, complicating integration into resource-constrained SoCs. To address these issues, researchers have explored adaptive and hybrid ramp generator designs. For instance, some solutions incorporate operational amplifiers combined with counters and digital-to-analog converters (DACs) to maintain ramp linearity, albeit at the cost of increased circuit complexity and area.

In this context, K. Paldurai et al. (2023) proposed three innovative ramp generation methods that significantly reduce complexity and silicon footprint while maintaining excellent linearity suitable for BIST architectures. The first method synthesizes a linear ramp by superimposing two sinusoidal signals of specific frequencies and phases. The non-linear sine

waves combine to create a linear segment suitable for ADC testing, leveraging the well-understood and stable characteristics of sinusoidal sources.

The second method involves generating a parabolic waveform by modulating a sine wave with a pulse signal, followed by differentiation to obtain a linear ramp. This approach creatively exploits waveform transformation and signal processing techniques to bypass the need for precision current sources.

The third and most promising approach utilizes a simple resistive charging scheme. Here, a fixed potential is maintained across a resistor, ensuring a constant charging current flows into a capacitor, thereby generating a highly linear voltage ramp. This technique is remarkable because it circumvents the challenges of conventional current sources, achieving near-ideal ramp linearity with minimal circuitry.

Simulations in a 90 nm CMOS process demonstrated that the resistive charging method reduces silicon area by up to 92% compared to previous approaches while achieving a DNL of only 0.0015 LSB when tested on an ideal 8-bit ADC. This performance marks a significant step toward integrating compact, low-power BIST modules in future ADC designs, facilitating in-field testing and calibration without expensive external equipment.

Parallel to efforts in ADC testing, the ADC design community has made substantial strides in hybrid architectures that aim to deliver high resolution and low power consumption. Among these, Noise-Shaping SAR ADCs (NS SAR ADCs) have emerged as a compelling solution that blends the low energy consumption of SAR ADCs with the enhanced resolution afforded by noise-shaping techniques typical of delta-sigma converters.

Traditional SAR ADCs use a binary search algorithm to successively approximate the input voltage. They are popular due to their moderate speed and energy efficiency but generally have limited resolution due to comparator noise and capacitor mismatch. Noise shaping, which involves pushing quantization noise out of the signal band, can improve effective resolution significantly when combined with oversampling and filtering.

Initial NS SAR ADC designs, such as the one proposed by J. Li and U.-K. Moon (2012), implemented a 10-bit, 100 MS/s converter with a two-stage loop filter comprising finite impulse response (FIR) and infinite impulse response (IIR) digital filters. Their architecture notably eliminated the need for a high-gain operational transconductance amplifier (OTA), which is typically power-hungry and area-expensive, thereby enhancing energy efficiency. The noise

shaping was achieved through a combination of loop filters and digital signal processing, demonstrating the feasibility of high-resolution SAR ADCs with noise shaping.

Subsequent work by Ali Badawy et al. introduced a dynamic amplifier-based loop filter that further reduced static power dissipation by avoiding continuous OTA biasing. Although effective in lowering power, this architecture required background calibration to mitigate process, voltage, and temperature (PVT) variations and nonlinearities introduced by the dynamic amplifier.

In 2018, J. Liu and B. Murmann explored digitally assisted noise shaping in SAR ADCs, leveraging a multi-stage modulator and digital feedback loops to enhance linearity and robustness. While these designs achieved greater flexibility and programmable noise shaping orders, they also increased digital logic complexity and silicon area.

To push resolution boundaries, M. Dong and R. Yan implemented a 12-bit NS SAR ADC with fourth-order noise shaping by cascading multiple FIR filters and using active amplification within the feedback loop. Their approach demonstrated high accuracy but increased power consumption and design complexity due to multiple amplification stages.

Recognizing these trade-offs, P. Harpe and G. Dolmans examined passive loop filters for NS SAR ADCs to reduce power and complexity. While passive filters consume less power, they are sensitive to PVT variations and exhibit lower robustness, limiting their applicability in variable environments.

Complementary architectures have also been explored. For instance, A. Zhan and M. Flynn integrated a noise-shaping SAR ADC within a delta-sigma feedback loop, highlighting the modular nature of noise shaping concepts. This approach provided design flexibility but emphasized the need for high-speed, low-noise comparators and amplifiers.

More recently, the development of novel amplifier topologies has become crucial in addressing the power, linearity, and area constraints inherent to NS SAR ADCs. Traditional high-gain OTAs, though precise, consume considerable power and silicon area and are susceptible to PVT variations, especially in deep-submicron processes. Inverter-based amplifiers have garnered attention due to their simplicity, low power consumption, and excellent PVT tolerance. These amplifiers operate in the inverter region, offering high gain with low supply voltage and reduced bias currents.

Leveraging inverter-based amplifiers within NS SAR ADCs promises to improve overall efficiency and integration density. Such approaches enable the design of ultra-low-power ADCs suitable for portable and battery-powered applications without compromising resolution or speed.

Collectively, the reviewed literature reflects a vibrant research landscape addressing both ADC testing and architecture optimization. In ramp generator design, transitioning from complex current source circuits to innovative waveform synthesis and simple resistive charging schemes offers a promising path toward compact, linear, and power-efficient BIST modules. These developments facilitate real-time ADC calibration and quality assurance within integrated systems.

Simultaneously, the evolution of NS SAR ADC architectures highlights the potential of hybrid analog-digital designs that exploit noise shaping to enhance resolution while managing power and area constraints. The ongoing exploration of amplifier topologies, loop filter designs, and digital assistance techniques demonstrates the nuanced trade-offs designers must balance.

Despite these advances, challenges remain. Integrating ultra-linear, low-area ramp generators into comprehensive BIST solutions for high-resolution ADCs still requires further refinement, particularly regarding noise immunity and process scalability. Similarly, NS SAR ADCs must continue evolving to improve robustness against PVT variations and simplify calibration, possibly through adaptive circuit techniques or emerging device technologies.

Future research will likely focus on co-optimizing test circuitry and ADC design, leveraging novel device physics, and exploiting digital calibration to achieve highly integrated, low-cost, and accurate ADC solutions for next-generation mixed-signal SoCs.

CHAPTER 4

TOOL DESCRIPTION

4.1 CADENCE

The Cadence tool package has been utilized for applications like schematic drawing, layout verification, and simulation. For various applications, multiple platforms are employed. Because it is open architecture, other vendor tools are also incorporated using a program called Design Framework II (DFW). The tools are given a common user interface and database. It is thus easy to switch between programs without changing the database.

In Cadence, graphical user interface is used with the views of DFW are:

- Command Interpreter Window (CIW) - controlling the environment.
- Design Window (DW) - the current design.
- Text Window (TW) - text is present.

THE DESIGN PROCESS

There are common structures for design tools. The libraries, views and instances are present in hierarchical form.

LIBRARIES AND VIEWS

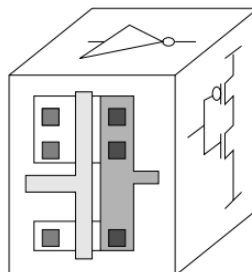


Figure 4. 1 An inverter cell with three views: layout, schematic, library

Library organization is applied for all design data in Cadence. Basic building blocks can be used in the construction and design of libraries are present in the reference libraries

section as shown in figure 4.1. There are different views and cells for each library as shown in the diagram below. Building block is formed using database objects called cells. Levels of abstraction of cells are represented as views which can be any of the schematic or layout or functional description.

INSTANCES AND HIERARCHY

The size and time of database is minimized by using hierarchical designs. The same type of 500 gates would be needed for a design. Then instead of building it 500 times, it is designed once and then used where it is needed. In this way one cell can be used (not copied) several times and each such use are called an instance of the cell. Every cell needs a symbol view that needs to be instantiated which represents a cell that it acts as handle. When a cell is instantiated, only the symbol is shown. Thus, by creating more complex structures by instantiating simple instances a hierarchical design is formed. Working on a selected level in a hierarchy, one can move up and down. By default, the highest one is the default when opened.

THE TECHNOLOGY FILE

Cadence must know for the technology to be used, the specifications (with different set of rules and properties), since there are different semiconductor processes. This information is stored which is a set of files existing on the system at different locations which is called Technology files. Specific technology will be connected when a library is created. The technology files contain information about:

- Layer definitions: Conductors, contacts, transistors
- Design rules: minimum size, distance to objects
- Display: Colours and patterns to use on the screen.
- Electrical properties: resistance, capacitance

Designer uses the technology files supplied by silicon vendor to fabricate the design with some standard cell libraries called as the Design Kit.

THE DESIGN FLOW

By creating the design in this step as shown in figure 4.2, drawing schematic views of all cells and blocks is done. The transistor symbols, resistors, capacitors and the wires which connects them are contained in the schematic view. On a schematic level, analyzing the design includes

functional verification (simulation) of the design. In a layout editor, layout is created which is the third step. By different colors, the final semiconductor layers are represented. On the final chip, all the cells and blocks used have the size. Verification is the last step of design.

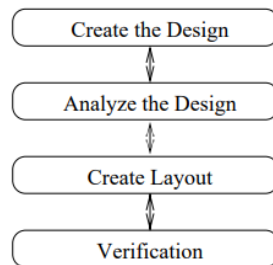


Figure 4.2 The design flow

SCHEMATIC AND SYMBOL TOOLS

Virtuoso Schematic Composer tool is used to create the schematic. By instantiating basic components and to connect them, interactive system editor is used for building schematics. The specifications, text and comments are suited by editing the values. The other parts of the construction can also use the symbols of the cells created by the editor.

SIMULATION

All the necessary netlists will be created, the simulation tool is started directly from the schematic editor. With the actual design included as an instance, in a test bench, which is also a schematic, a simulation is usually performed. The signal sources and power supply are also included in the test bench. the design for a wide range of variables can be quickly analyzed by using parameters for the properties of the components. Affirm Analog Circuit Design Environment within which the simulator is run to handle the interface between the user and the simulator. A wide range of analyzes which can be saved and presented graphically is offered by simulator. Various parameters of circuit analyzed - delay time, rise time, slew rate, phase margin is done by feeding the results into a calculator.

LAYOUT TOOL

For drawing the layout, the Virtuoso Layout Editor is used. Different color geometric figures are present in a layout. From the size and color of this-figures to generate the final

mask layers used in the design fabrication. By instantiating their layout views, other cells can be included. To verify that the layout fulfils all electrical and geometric rules a Design Rule Check (DRC) program is used. Error will be marked in the design by the tool so it can be simulated.

PLACE AND ROUTE

Place and route are called the construction of a large design which is the final stage. All the components of chip are placed on its locations and connected to each other at this stage. Manually it takes a lot of time and is tedious to design which consist of thousands of connection points. To place the components, output buffers, memory structures, amplifiers, various alternatives are tried by the designer. Envisia Silicon Ensemble, later the place and route tool that will be described. At the same time, it is a very important program a very large design while respecting some design constraints (restrictions on delay and size) that that can place and route. For Standard Cell designs, Silicon Ensemble is used but it can handle other structures as well. To run Silicon Ensemble as a standalone tool, Cadence manual describes since not all designs that is to be routed are created.

CHAPTER 5

PROPOSED METHODOLOGY

5.1 Fully Differential OTA-Based Loop Filter Design

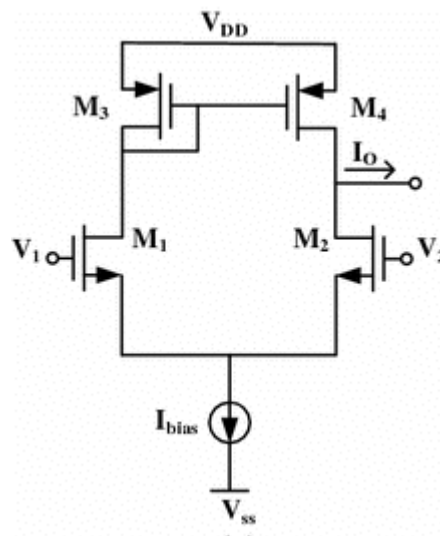


Figure 5.1 Basic Differential OTA Filter

Operational Transconductance Amplifier (OTA) based filters play a crucial role in high-resolution ADC systems such as Noise-Shaping SAR ADCs. The accuracy and linearity of the filter directly influence the spectral shaping of quantization and comparator noise. However, conventional OTA filters often suffer from process, voltage, and temperature (PVT) variations and non-idealities like finite gain and poor common-mode rejection, all of which degrade the Signal-to-Noise and Distortion Ratio (SNDR). The proposed OTA filter addresses these issues by employing a fully differential topology with enhanced linearity, improved PVT tolerance, and tunable filter characteristics.

The primary role of this filter is to act as a residue integrator in a loop filter configuration (such as an IIR filter in noise-shaping SAR ADCs). The differential OTA topology benefits from better immunity to common-mode noise and harmonic distortion, making it suitable for advanced, low-voltage CMOS nodes.

A simplified expression for the integrator's output, assuming ideal OTA behavior, is:

$$V_{out}(s) = \frac{1}{sRC} V_{in}(s)$$

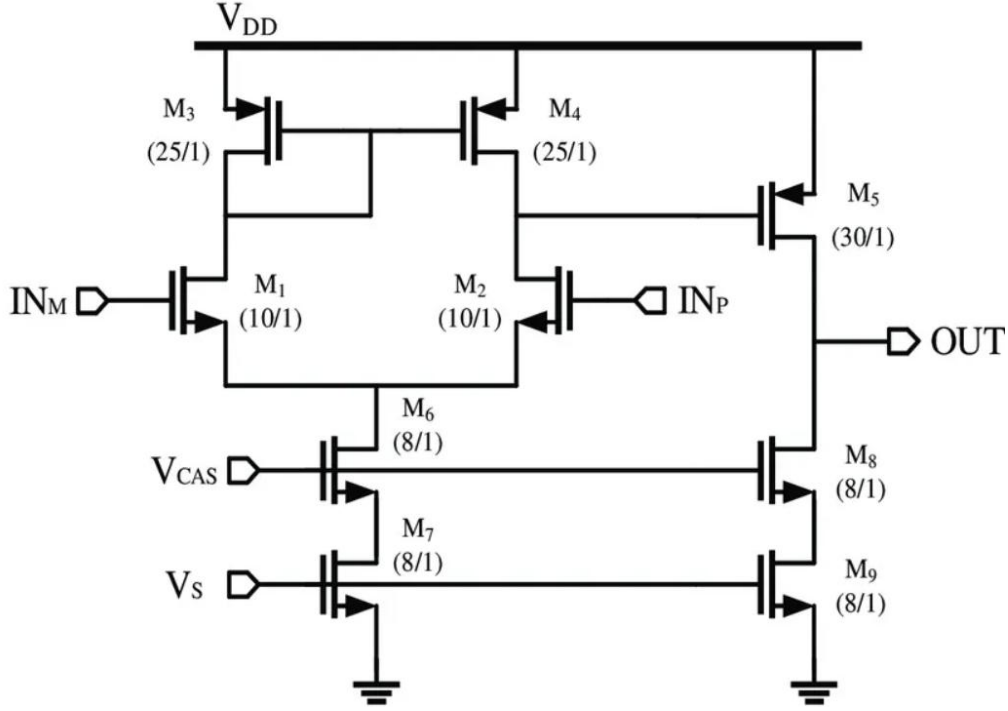


Figure 5.2 Fully Differential Operational Transconductance Amplifier (OTA) filter

Where R and C are the effective resistance and capacitance in the feedback network. In practical designs, resistors (e.g., source degeneration) and capacitors are combined with the OTA to control bandwidth and gain, tuning the loop filter's Noise Transfer Function (NTF).

5.1.1 CIRCUIT IMPLEMENTATION

The proposed Operational Transconductance Amplifier (OTA) filter, depicted in Figure 5.3, is designed using a fully differential, inverter-based architecture that incorporates both capacitive and resistive feedback elements to achieve desired filtering characteristics. At its core, the OTA employs a pseudo-differential configuration by combining NMOS and PMOS transistors (specifically, M6 and M10) as the main differential pair. This complementary arrangement enhances the transconductance of the amplifier, allowing it to operate with higher gain and improved efficiency. The differential nature of the architecture not only improves common-mode rejection but also supports high linearity and better signal swing. The load of

the OTA is implemented using PMOS transistors M7 and M8, which are actively regulated through a common-mode feedback (CMFB) circuit. This CMFB loop is crucial for maintaining a stable output common-mode voltage, ensuring reliable and symmetrical output signal operation, which is essential in fully differential systems. To control the operating current and set the quiescent point of the amplifier, a tail current source (M9) is used. This transistor defines the bias current, ensuring the OTA operates within the desired power envelope.

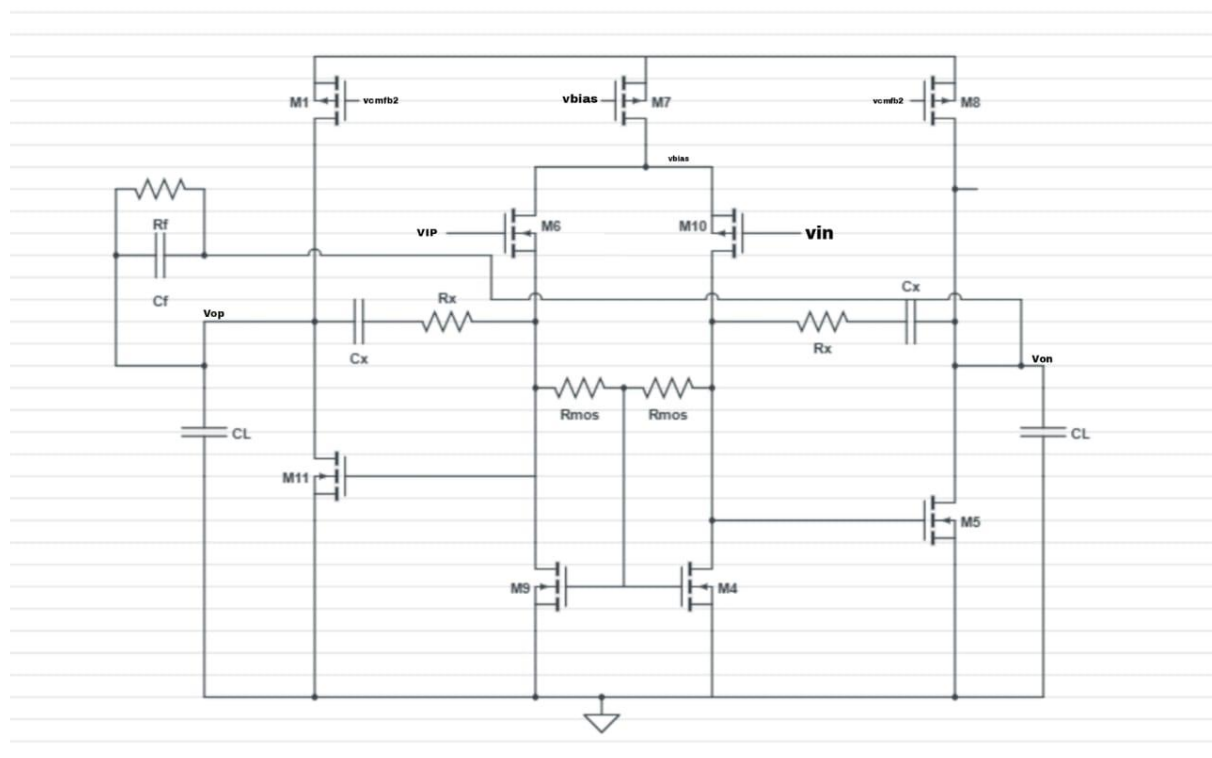


Figure 5.3 Proposed Operational Transconductance Amplifier (OTA) filter

To improve linearity and reduce distortion—especially important in high-resolution analog signal processing—a resistive degeneration element, typically implemented as a MOS resistor (R_{mos}), is included. This element provides local feedback, effectively moderating the transconductance of the input pair and making the amplifier's behavior more linear over a wider range of input voltages. The frequency response of the filter is shaped using a combination of feedback capacitors (C_f) and resistors (R_f). These components determine whether the OTA behaves as an integrator or a low-pass filter, depending on their chosen values. By appropriately tuning these parameters, designers can target specific bandwidths and filter slopes, allowing the circuit to adapt to various signal conditioning tasks. Additionally, the filter includes series components at the input—namely R_x and C_x —that form a feed-forward path. This configuration mimics the behavior of finite impulse response (FIR) filters,

enhancing the filter's selectivity and frequency response control without relying solely on feedback. Overall, the proposed OTA filter combines the benefits of high transconductance, enhanced linearity, stable common-mode control, and flexible frequency shaping to support efficient analog signal processing in a compact, low-power implementation.

LINEARITY AND STABILITY ANALYSIS

To enhance linearity and stability, differential output capacitors (**CL**) are employed to suppress high-frequency peaking. The feedback loop ensures that the OTA operates in a closed-loop configuration, which minimizes sensitivity to transistor mismatch and PVT drift. Mathematically, the loop gain **A_{loop}** of the system is given by:

$$A_{loop}(s) = A_{OTA} \cdot \frac{Z_f(s)}{Z_{in}(s)}$$

Where $Z_f(s)$ and $Z_{in}(s)$ are the feedback and input impedances respectively. The high open-loop gain of the OTA ensures accurate integration with minimal gain error.

COMMON-MODE FEEDBACK (CMFB) STRATEGY

The circuit incorporates a Common-Mode Feedback (CMFB) loop, a crucial mechanism designed to maintain the stability of the output common-mode voltage in fully differential amplifier architectures. In this configuration, the CMFB circuit continuously monitors the average voltage of the differential outputs, denoted as V_{op} and V_{on} . By sensing this average, the CMFB loop actively adjusts the gate voltages of the PMOS load transistors (M7 and M8), thereby regulating the output common-mode level. This regulation is essential for ensuring that the amplifier operates within its intended range, particularly when dealing with supply voltage fluctuations, process variations, or mismatches between transistors. The CMFB loop effectively prevents the common-mode voltage from drifting during dynamic signal operation, which could otherwise lead to degraded performance or distortion.

In modern CMOS technologies, where the supply voltage is scaled down to minimize power consumption—often to values as low as 0.9V—the margin for error in maintaining the

output common-mode level becomes significantly smaller. Without the presence of a CMFB circuit, the output signal swing could become severely limited or even unstable, compromising the linearity and dynamic range of the amplifier. Additionally, a drifting common-mode voltage could push the output transistors into non-ideal operating regions, leading to distortion or clipping. Thus, the CMFB loop plays a vital role not only in preserving the integrity of the differential signal but also in enabling reliable operation in low-voltage, scaled CMOS environments. Its inclusion ensures that the amplifier delivers consistent performance, high linearity, and full utilization of the available output swing, all of which are critical in precision analog and mixed-signal circuit applications.

BENEFITS OF THE PROPOSED DESIGN

- **PVT Robustness:** Differential architecture with common-mode feedback offers strong immunity to process and temperature shifts.
- **High Linearity:** Source degeneration resistors reduce harmonic distortion, making the filter suitable for high-precision applications.
- **Scalability:** Inverter-based OTA design is compatible with advanced CMOS processes, offering compact layout and low power.
- **Programmability:** Filter coefficients (R_f , C_f , R_x , C_x) can be tuned digitally or via trimming to set NTF characteristics.

5.2 FILTER BASED SAR ADC

The proposed solution is an 8-bit Noise-Shaping Successive Approximation Register (NS SAR) Analog-to-Digital Converter (ADC) that integrates a loop filter and a dynamic comparator to achieve high resolution, improved linearity, and reduced power consumption. The architecture is designed for low-power applications such as biosensors, IoT devices, and portable electronics. The block diagram of Filter based SAR ADC is shown in figure 5.4. Below is a detailed explanation of the proposed solution:

The input signal is first sampled and held using an input bootstrapping circuit as shown in figure 5.5, which acts as the sample-and-hold (S/H) stage. This circuit ensures accurate sampling of the input signal by minimizing signal distortion and charge injection. The bootstrapping technique enhances the linearity of the sampling process, making it suitable for

high-resolution applications. The sampled signal is then passed to the Capacitive Digital-to-Analog Converter (CDAC) for further processing.

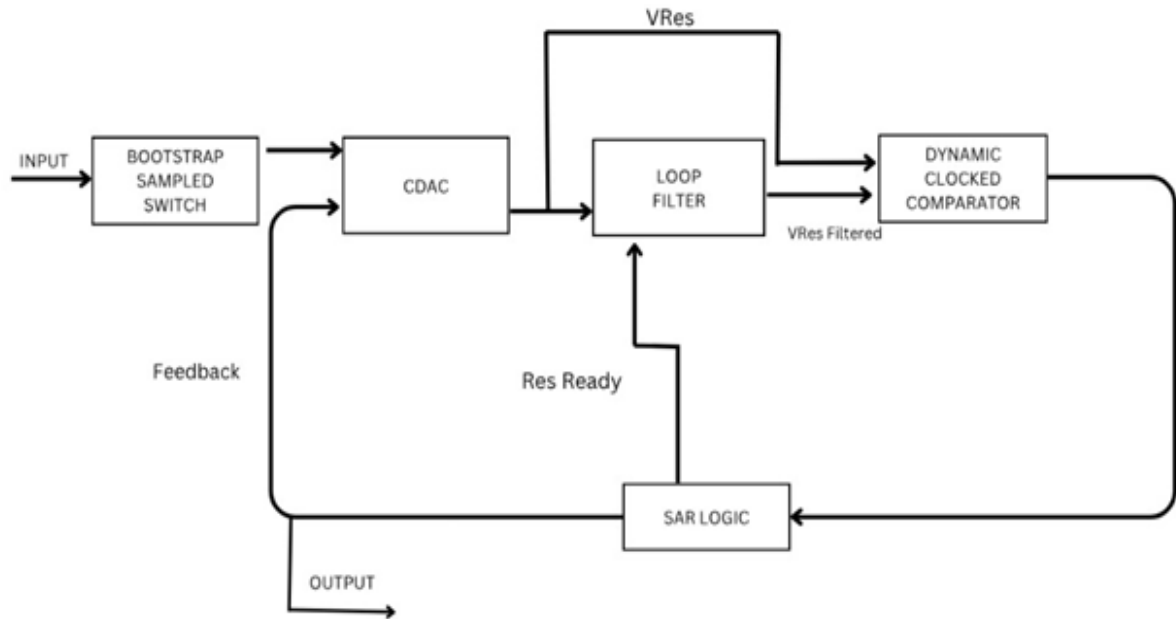


Figure 5.4 Flowchart of Filter Based SAR ADC

The CDAC as shown in figure 5.6 converts the sampled analog signal into a digital output based on a reference voltage (V_{ref}) and the previous DAC output. The CDAC operates under the control of the SAR logic, which determines the bit decisions during the successive approximation process.

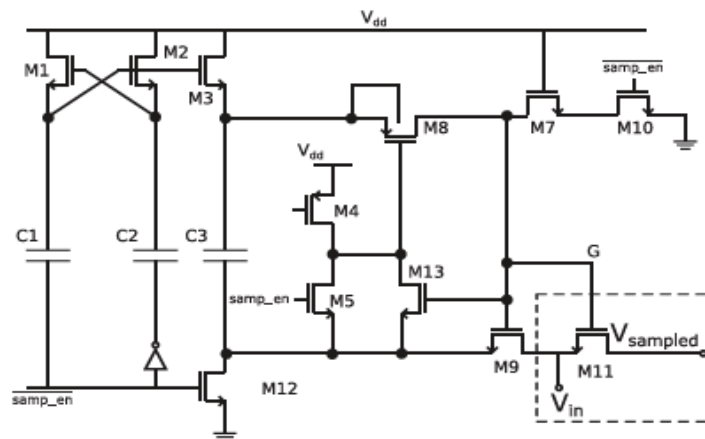


Figure 5.5 Input Bootstrap Sampling Switch

The CDAC also calculates the residual voltage (V_{Res}), which represents the difference between the input signal and the quantized output. This residual voltage is critical for noise shaping and is passed to the loop filter for further processing.

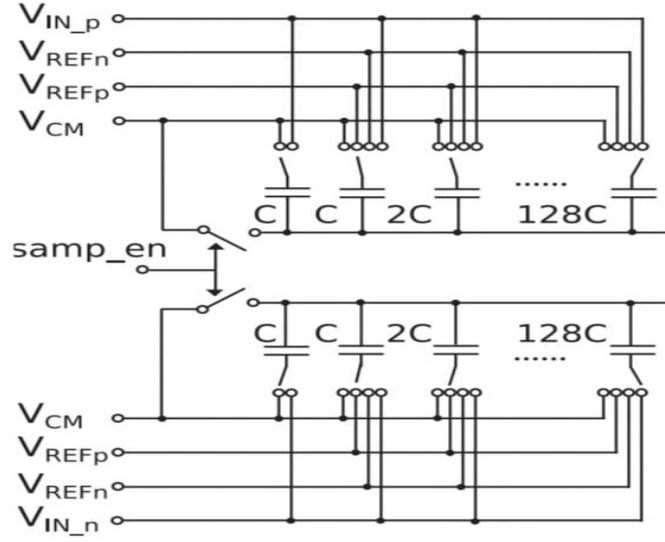


Figure 5.6 Capacitive Based DAC

The residual voltage produced by the Capacitive Digital-to-Analog Converter (CDAC) serves as a critical signal in the noise-shaping operation of the ADC. This voltage, representing the difference between the input signal and the current DAC output, is passed to a dedicated loop filter. The primary role of this loop filter is to shape the spectral distribution of quantization noise, effectively pushing it to higher frequencies outside the band of interest. By doing so, the filter significantly improves the in-band signal fidelity and enhances the overall resolution and signal-to-noise ratio (SNR) of the ADC.

The loop filter is designed using an analog integrator configuration, employing low-leakage capacitors and carefully biased operational transconductance amplifiers (OTAs). This configuration ensures high linearity and efficient suppression of noise components while maintaining low power consumption. Moreover, the design is optimized to remain stable and accurate under varying process, voltage, and temperature (PVT) conditions, which is essential for consistent performance in real-world applications.

Once the residual voltage is filtered, the resulting signal ($V_{\text{Res-Filtered}}$) is directed to a high-speed dynamic comparator. This comparator is responsible for performing bit-wise voltage comparisons between the filtered residual signal and the output of the CDAC. Implemented using a regenerative latch architecture, the comparator offers rapid decision-making with minimal static power dissipation. It is designed to be sensitive to small voltage differences, ensuring accurate bit decisions even in low-voltage conditions.

The output of the comparator is forwarded to the Successive Approximation Register (SAR) logic. The SAR logic orchestrates the conversion process by sequentially evaluating each bit of the digital output based on the comparator's feedback. It updates the CDAC control bits in each cycle, refining the approximation of the input signal. The SAR logic is optimized for high-speed operation and minimal switching activity, contributing to the overall energy efficiency of the ADC.

The iterative interaction between the SAR logic and CDAC forms a closed-loop system that continuously minimizes quantization error over multiple conversion cycles. This feedback mechanism is essential for implementing noise shaping and ensures that the digital output converges to a high-accuracy representation of the input signal. The combination of the CDAC, loop filter, dynamic comparator, and SAR logic allows the proposed ADC architecture to achieve high resolution, power efficiency, and robustness suitable for applications such as biomedical sensing, IoT edge devices, and portable electronics.

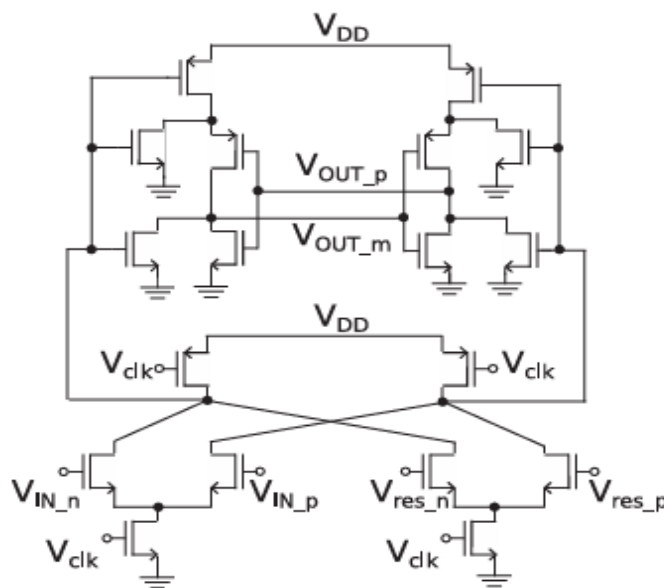


Figure 5.7 Dynamic Clocked Comparator

The ADC output is fed back to the CDAC, completing the noise-shaping loop. This feedback mechanism ensures that the quantization noise is shaped away from the signal band, improving the overall resolution and signal-to-noise ratio (SNR). The feedback loop also enhances the stability and robustness of the ADC, making it suitable for high-resolution applications.

5.2.1 CIRCUIT IMPLEMENTATION:

The proposed 8-bit Noise-Shaping SAR ADC was implemented using the Cadence Virtuoso design environment in 90nm CMOS technology. The entire architecture was modularly designed and each block was verified for functionality, low-power operation, and compatibility with noise-shaping principles.

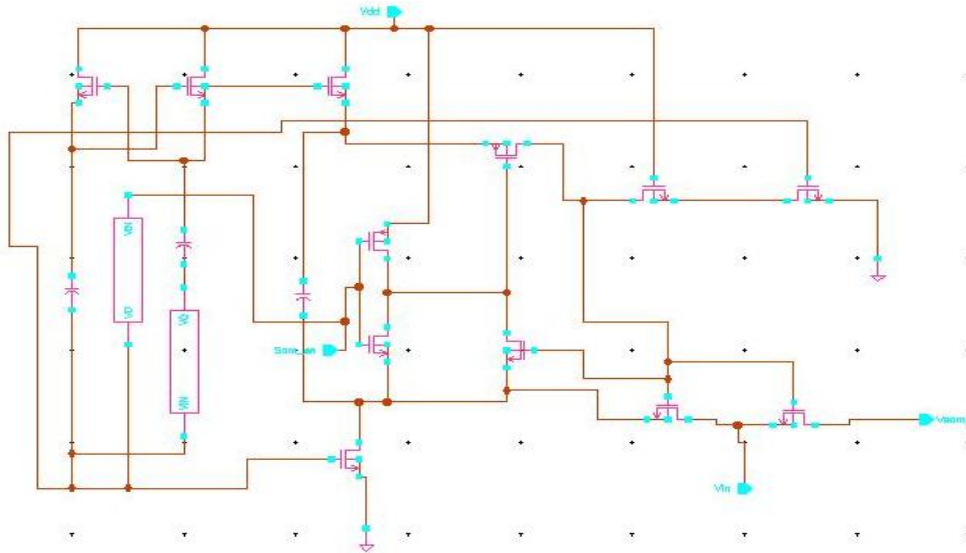


Figure 5.8 Circuit Implementation of Input Bootstrap Sampling Switch

The implementation emphasizes accuracy, power efficiency, and robustness against PVT (process-voltage-temperature) variations, making it suitable for integration in portable and biomedical applications.

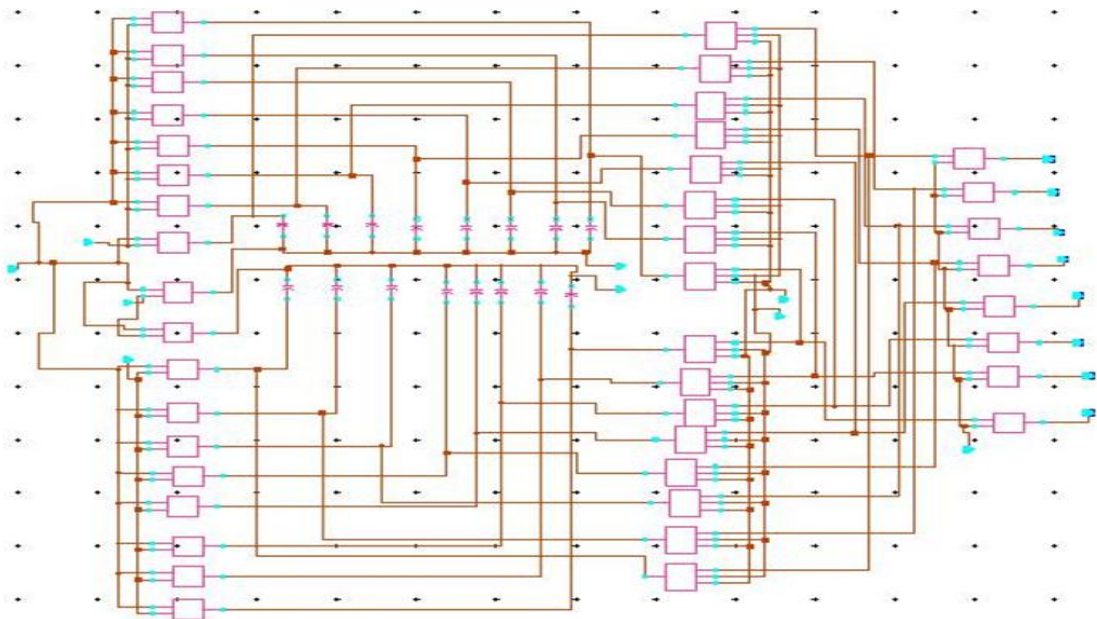


Figure 5.9 Circuit Implementation of Capacitive Based DAC

The input bootstrapping switch as shown in figure 5.8 serves as the sample-and-hold stage in the ADC architecture. This circuit was implemented using a transmission gate configuration with bootstrapping to ensure high linearity and minimal signal distortion. The bootstrap circuit maintains a constant gate-to-source voltage (V_{gs}) across the sampling switch during the switching phase. Simulation results confirmed accurate tracking and high linearity across the input voltage range, which is essential for high-resolution conversion.

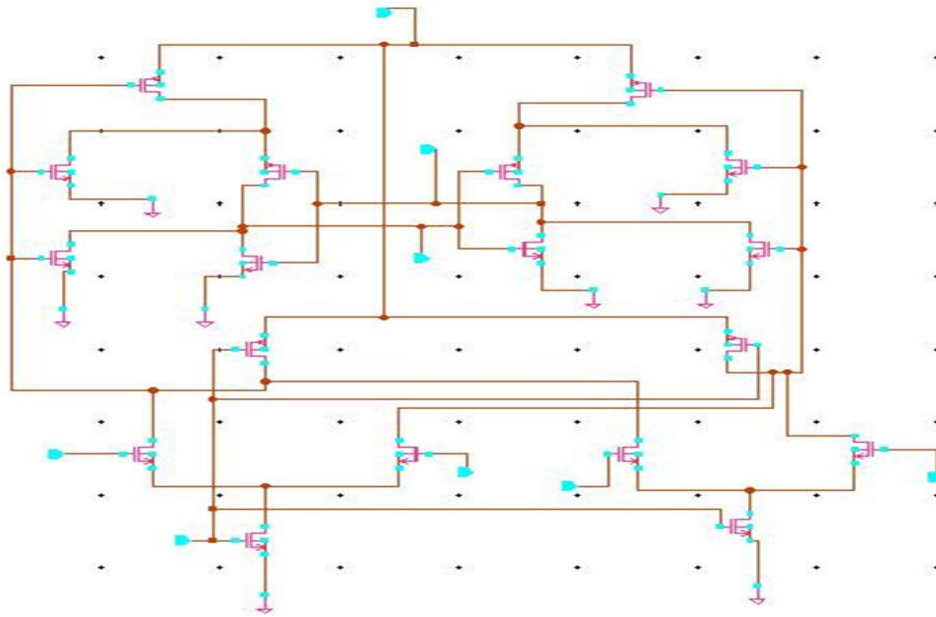


Figure 5.10 Circuit Implementation of Dynamic Clocked Comparator

The Capacitive Digital-to-Analog Converter (CDAC) as shown in figure 5.9 was implemented using a binary-weighted capacitor array. A top-plate sampling architecture was employed to minimize switching power and enhance linearity.

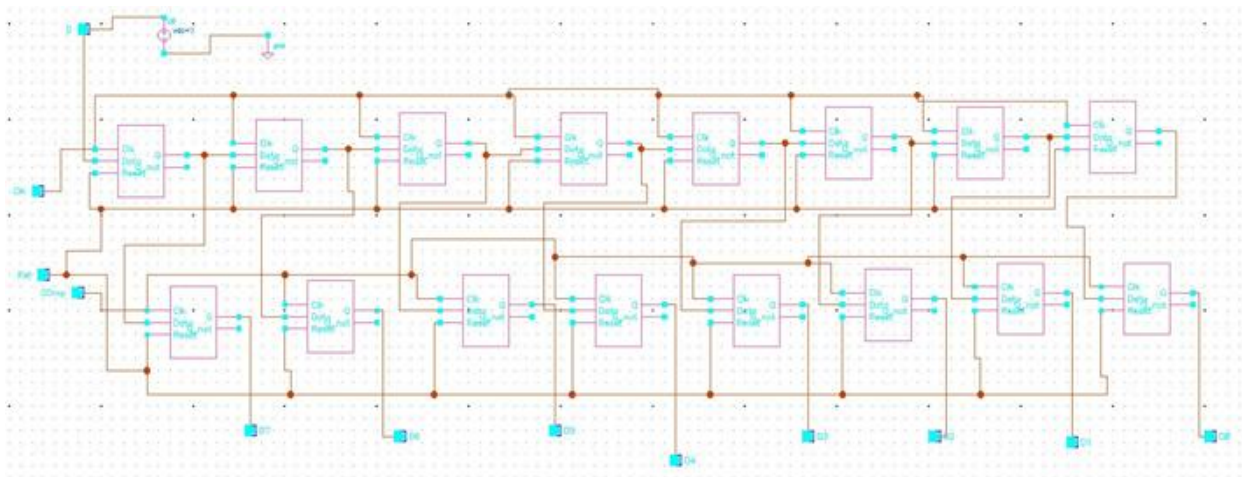


Figure 5.11 Circuit Implementation of SAR Logic

The Capacitive Digital-to-Analog Converter (CDAC) serves a dual function in the proposed architecture: performing digital-to-analog conversion and generating the residual voltage necessary for noise shaping. This residual voltage (V_{Res}) is obtained by subtracting the DAC output from the input signal and is critical to the operation of the noise-shaping loop. The reference voltage switching logic within the CDAC has been optimized to reduce dynamic power consumption during the successive approximation process.

To facilitate noise shaping, a loop filter is implemented as an analog integrator. This integrator processes the residual voltage and spectrally pushes quantization noise out of the signal band, thereby enhancing the effective resolution of the ADC. The filter utilizes low-leakage capacitors and carefully biased operational transconductance amplifiers to ensure low power consumption and stable performance across process, voltage, and temperature (PVT) variations. The filtered residual voltage ($V_{\text{Res-Filtered}}$) is then passed to the comparator stage for quantization.

Unlike designs that rely on a ramp generator to provide a timing reference for the comparator, the proposed implementation employs an explicitly provided clock signal. This approach simplifies the overall design, reduces circuit complexity, and minimizes power consumption while ensuring accurate and synchronized operation of the dynamic comparator.

The dynamic comparator, illustrated in Figure 5.10, is based on a regenerative latch architecture. It compares the filtered residual voltage from the loop filter with the CDAC output and generates a digital decision accordingly. This output is used by the SAR logic to determine each bit during the conversion process. The comparator is optimized for high speed and low power, featuring fast regeneration, minimal static current consumption, and robust noise and offset performance to ensure reliable bit decisions.

Finally, the SAR logic as shown in figure 5.11 was implemented using a synchronous finite state machine. It controls the operation of the CDAC, manages the bit-by-bit approximation process, and coordinates with the ramp generator to ensure accurate timing. The SAR logic was optimized for low switching activity to minimize dynamic power consumption. The feedback from the SAR logic to the CDAC closes the noise-shaping loop and refines the quantization with each cycle, enhancing resolution and dynamic range. The Complete Circuit implementation of filter based SAR ADC is shown in figure 5.12.

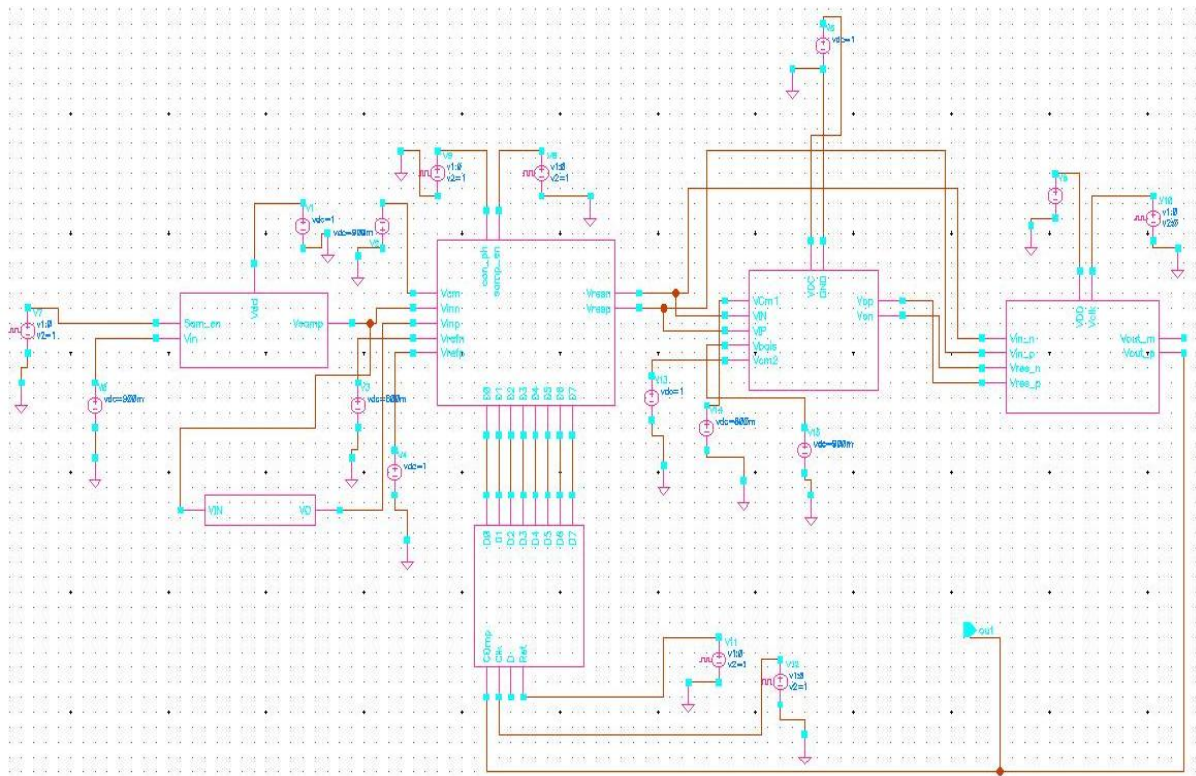


Figure 5.12 Circuit Implementation of Filter based SAR ADC

Each of these blocks was individually simulated and verified under different operating conditions. Layouts were created with careful attention to parasitic matching and area optimization. Post-layout simulations further confirmed that the design met the required specifications for resolution, power, and linearity. Overall, the proposed NS SAR ADC architecture, implemented in Cadence with 90nm technology, demonstrates a high-performance, energy-efficient solution suitable for modern low-power analog-to-digital conversion applications.

CHAPTER 6

EXPERIMENT AND RESULTS

6.1 OTA-BASED LOOP FILTER FOR NS-SAR ADC

The OTA-based loop filter plays a critical role in shaping quantization and comparator noise in the Noise-Shaping SAR ADC. The designed filter is a fully differential, inverter-based OTA with resistive-capacitive feedback that acts as a continuous-time integrator (IIR stage).

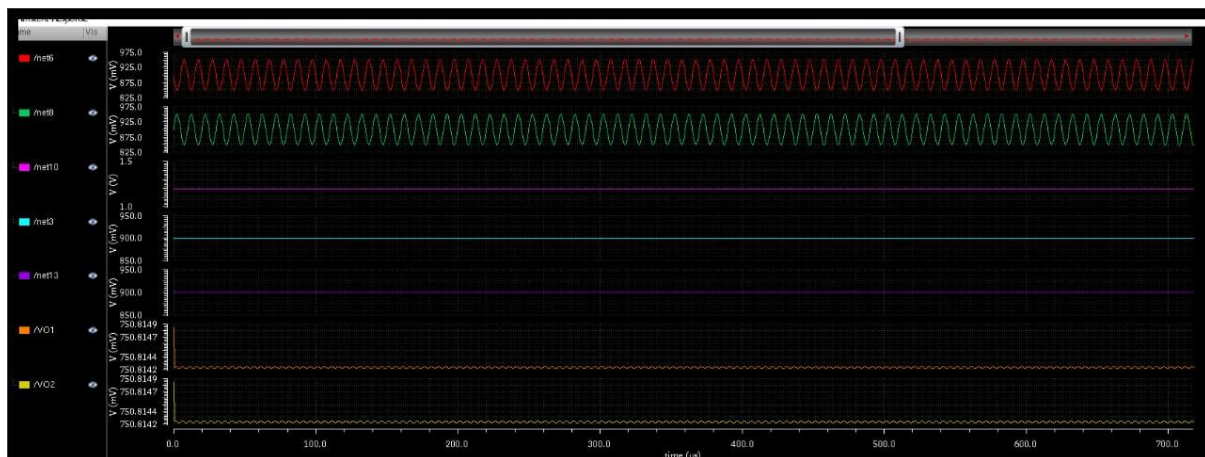


Figure 6.1 OTA Filter Output Waveform

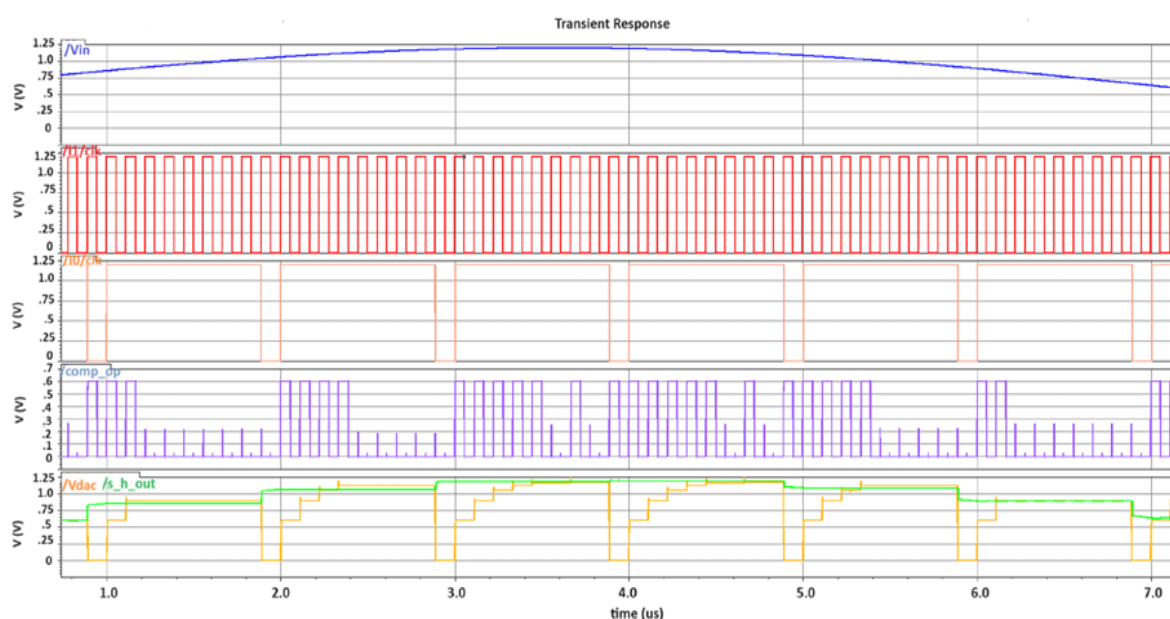


Figure 6.2 Output of Filter Based Noise Shaping SAR ADC

The Circuit is implemented in Cadence Virtuoso using GPDK 90nm CMOS technology. The transient simulation of the OTA filter was performed to verify the integration behavior and assess the filter's impact on the in-band noise. The OTA output in response to SAR residue inputs is shown in Figure 6.1. The Output Waveform of the Filter based Noise Shaping SAR ADC is shown in Figure 6.2.

6.1.1 COMPARISON WITH EXISTING RESULTS

Parameter	This Work	[20] Fredenburg JSSC'12	[18] Liu ISSCC'17	[16] Tang ISSCC'20	[22] Jie ISSCC'20	[24] Liu ISSCC'20	[21] Lin ISSCC'19	[17] Jie ISSCC'19	[9] Yi TCASII'22
Technology (nm)	90	65	28	40	28	40	14	40	65
Supply (V)	1.2	1.2	1	1.1	1	1.1	0.9	1	1.2
OSR	4	4	13.2	8	10	25	4	4	16
BW (MHz)	11.25	11	5	0.625	0.1	0.04	40	50	0.625
Power (mW)	0.62	0.81	0.46	0.107	0.12	0.0674	1.25	13	0.113
Sharp NTF	Yes	Yes	No	Yes	Yes		No	No	Yes
Fully Dynamic	No	No	Yes	Yes	No	Yes Partially	Yes Partially	No	No
PVT Robust	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes
Active Gain Stages	1	1	2	2	2	0	0	1	1
Residue Processing	OTA (Inverter-Based)	Static SS	Dynamic SS	2 DA	2 Static Amp	Passive	Passive	Static SS	Static SS
Gain Loop	Closed-loop	Open-loop	Closed-loop	Closed-loop	Closed-loop	—		Open-loop	Open-loop

Table 6.1 Summary of the recently published SAR ADC performance parameters

Table 6.1 presents a comprehensive comparison of recently published SAR ADCs with respect to key performance metrics such as CMOS process node, supply voltage, power consumption, silicon area, sampling frequency (F_s), resolution, Effective Number of Bits (ENOB), Signal-to-Noise-and-Distortion Ratio (SNDR), and Spurious-Free Dynamic Range (SFDR). The data serves as a benchmark to evaluate the advancements in SAR ADC design across various trade-offs such as speed, resolution, power, and integration area. It has been implemented in 90nm technology and its layout shows a minimum area of 0.000376 mm^2 .

CHAPTER 7

CONCLUSION

The proposed method presents a noise-shaping-based enhancement to the Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) architecture, aiming to achieve high resolution, low power consumption, and improved linearity. A key innovation in this work is the integration of a loop filter that spectrally shapes quantization noise, thereby pushing unwanted noise components out of the signal band and enhancing the in-band signal quality.

The architecture leverages a Capacitive Digital-to-Analog Converter (CDAC) for both digital-to-analog conversion and residual voltage generation, which serves as the input to the noise-shaping filter. This loop filter is implemented as a continuous-time integrator using fully differential OTA-based circuits that ensure low distortion, high linearity, and strong immunity to process-voltage-temperature (PVT) variations. These characteristics are vital for maintaining consistent performance across diverse operating conditions.

The filtered residual voltage is processed using a dynamic comparator, which performs high-speed comparisons with low power consumption. Controlled by explicitly timed clock signals, the comparator delivers accurate decision-making necessary for the SAR logic to iteratively determine the digital output. The SAR logic itself is optimized for energy efficiency and speed, enabling the proposed ADC to operate effectively in resource-constrained environments.

This feedback-based architecture not only improves accuracy but also suppresses non-linearity errors, leading to a marked improvement in key performance metrics such as Differential Non-Linearity (DNL) and Integral Non-Linearity (INL). Simulation results validate the effectiveness of the design in achieving high resolution and noise resilience.

Fabricated using 90 nm CMOS technology, the ADC demonstrates compact area utilization, making it well-suited for integration into system-on-chip (SoC) platforms. Its low power profile and robust performance make it ideal for a wide range

of modern electronic applications, including biomedical devices, IoT sensor nodes, and portable consumer electronics.

In summary, the proposed noise-shaping SAR ADC design offers a balanced and efficient solution that addresses the critical requirements of precision, power efficiency, and scalability. Its modular architecture and compatibility with advanced CMOS processes position it as a promising candidate for next-generation mixed-signal systems.

BIBLIOGRAPHY

- [1] Badawy, A. and Ismail, A., 2024. Inverter-based noise-shaping SAR ADC for low-power applications. *Microelectronics Journal*, 144, p.106094
- [2] K. Paldurai, K. Hariharan, K.S. Naveen et al., Implementation of Linear Test Stimulus Generator for Non-linearity Computation of ADC, *Instrum Exp Tech*, vol. 66, pp. 570–577, 2023.
- [3] Q. Zhang, J. Li, N. Ning, Q. Yu, K. Wu, Z. Zhang, "A second-order noise shaping SAR ADC with error-feedback structure and data weighted averaging," *Microelectron. J.*, vol. 105, 2020.
- [4] Y. Li, Y. Zhao, Y. Zhao, M. Ye, A 12.1bit-ENOB noise shaping SAR ADC for biosensor applications, *Microelectron. J.*, vol. 118, 2021.
- [5] Liu, X. Wang, Z. Gao, M. Zhan, X. Tang, N. Sun, 9.3 A 40 kHz-BW 90 dB-SNDR noise-shaping SAR with 4x passive gain and 2nd-Order mismatch error shaping, in *2020 IEEE International Solid-State Circuits Conference (ISSCC)*, 2020, pp. 158–160.
- [6] A. Ismail, I. Mostafa, A process-tolerant, low-voltage, inverter-based OTA for continuous-time sigma-delta ADC, *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 24, no. 9, pp. 2911–2917, 2016.
- [7] Elganimi, T. Y. (2014, September). Design and analysis of a bootstrap ramp generator circuit based on a bipolar junction transistor (BJT) differential pair amplifier. In *Proceedings of the World Congress on Engineering and Computer Science* (Vol. 1).
- [8] Azaïs, F., Bernard, S., Bertrand, Y., & Renovell, M. (2002). Analog built-in saw-tooth generator for ADC histogram test. *Microelectronics journal*, 33(10), 781-789.
- [9] P. Yi, Y. Liang, S. Liu, N. Xu, L. Fang, Y. Hao, A 625 kHz-BW, 79.3 dB SNDR second-order noise-shaping SAR ADC using high-efficiency error-feedback structure, *IEEE Trans. Circuits Syst. II* 69 (3) (2022) 859–863.
- [10] S. Li, B. Qiao, M. Gandara, N. Sun, A 13-ENOB 2nd-order noise-shaping SAR ADC realizing optimized NTF zeros using an error-feedback structure, in: *2018 IEEE International Solid- State Circuits Conference- (ISSCC)*, 2018, pp. 234–236.
- [11] Tasneem NT, Mahbub I (2021) A 2.53 NEF 8-bit 10 kS/s 0.5 μm CMOS neural recording read-out circuit with high linearity for neuromodulation implants. *Electronics* 10(5):1–15
- [12] Chaput S, Brooks D, Wei GY (2018) An area-efficient 8-bit single-ended ADC with extended input voltage range. *IEEE Trans Circuits Syst II Express Briefs* 65(11):1549–1553

- [13] Savitha M, Reddy RVS (2019) Dual split-three segment capacitor array design based successive approximation ADC for Io-T ecosystem. *Integration* 69:279–288
- [14] L. Jie, B. Zheng, M.P. Flynn, 20.3 A 50 MHz-bandwidth 70.4 dB-SNDR calibration-free time-interleaved 4th-Order noise-shaping SAR ADC, in: 2019 IEEE International Solid- State Circuits Conference- (ISSCC), 2019, pp. 332–334.
- [15] A. Abo, P. Gray, A 1.5-V, 10-bit, 14.3-MS/s CMOS pipeline analog-to-digital converter, *IEEE J. Solid-State Circuits* 34 (5) (1999) 599–606.
- [16] X. Tang, X. Yang, W. Zhao, C.-K. Hsu, J. Liu, L. Shen, A. Mukherjee, W. Shi, D.Z. Pan, N. Sun, 9.5 A 13.5b-ENOB second-order noise-shaping SAR with PVT-robust closed-loop dynamic amplifier, in: 2020 IEEE International Solid- State Circuits Conference- (ISSCC), 2020, pp. 162–164.
- [17] Bae S, Lee S, Seong S, Kong S, Park B, Lee M (2023) A 13-bit 1-MS/s SAR ADC with completion-aware background capacitor mismatch calibration. *IEEE Access* 11:104323–104332
- [18] Banik S, Rasel MMH, Mahmud T, Hasanuzzaman M (2020) Design and implementation of a low-power 1V, 77.26 μ W 6-bit SAR ADC in Cadence 90 nm CMOS process for biomedical application. In: IEEE region 10 symposium (TEN SYMP), pp 839–842, Dhaka, Bangladesh
- [19] Fahmy GA, Zorkany M (2021) Design of a memristor-based digital to analog converter (DAC). *Electronics* 10(5):1–17
- [20] L. Jie, B. Zheng, H.-W. Chen, R. Wang, M.P. Flynn, 9.4 A 4th-order cascaded noise-shaping SAR ADC with 88 dB SNDR over 100 kHz bandwidth, in: 2020 IEEE International Solid- State Circuits Conference- (ISSCC), 2020, pp. 160–162.
- [21] Y.-Z. Lin, C.-Y. Lin, S.-C. Tsou, C.-H. Tsai, C.-H. Lu, 20.2 A 40 MHz-BW 320 MS/s passive noise-shaping SAR ADC with passive signal-residue summation in 14 nm FinFET, in: 2019 IEEE International Solid- State Circuits Conference (ISSCC), 2019, pp. 330–332.
- [22] J. Liu, X. Wang, Z. Gao, M. Zhan, X. Tang, N. Sun, 9.3 A 40 kHz-BW 90 dB-SNDR noise-shaping SAR with 4x passive gain and 2nd-Order mismatch error shaping, in: 2020 IEEE International Solid- State Circuits Conference- (ISSCC), 2020, pp. 158–160.
- [23] L. Meng, J. Chen, M. Zhao, Z. Tan, An 18.2 μ W 101.1 dB DR fully-dynamic $\Sigma\Delta$ ADC with partially-feedback noise-shaping quantizer and CLS-embedded two stage FIAs, in: ESSCIRC 2023- IEEE 49th European Solid State Circuits Conference (ESSCIRC), 2023, pp. 393–396.
- [24] Mohammedali, M.A. and Al-Gayem, Q., 2023. An Accurate and Fast Method for Improving ADC Nonlinearity. *Applied Computational Intelligence and Soft Computing*, 2023(1), p.8899666.

- [25] Y.-S. Shu, L.-T. Kuo, T.-Y. Lo, An oversampling SAR ADC with DAC mismatch error shaping achieving 105 dB SFDR and 101 dB SNDR over 1 kHz BW in 55 nm CMOS, *IEEE J. Solid-State Circuits* 51 (12) (2016) 2928–2940.
- [26] Z. Chen, M. Miyahara, A. Matsuzawa, A 9.35-ENOB, 14.8 fJ/conv.-step fully-passive noise-shaping SAR ADC, *IEICE Trans. Electron.* 99 (8) (2016) 963–973.
- [27] Kuo HL, Lu CW, Chen P (2021) An 18.39 fJ/conversion-step 1-MS/s 12-bit SAR ADC with non-binary multiple-LSB redundant and non-integer-and-split-capacitor DAC. *IEEE Access* 9:5651–5669

